

Computations in low-dimensional topology: quantum knot polynomials and ribbon concordance

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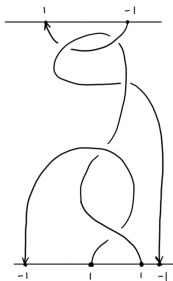
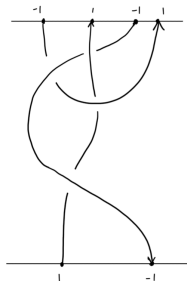
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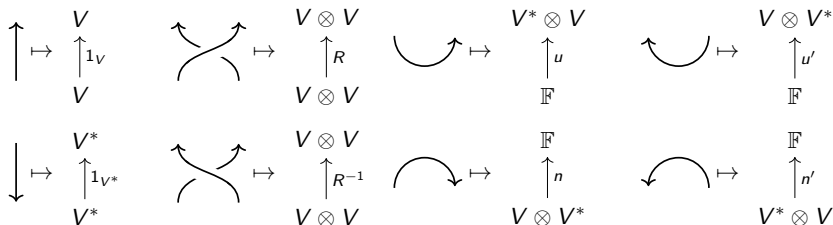
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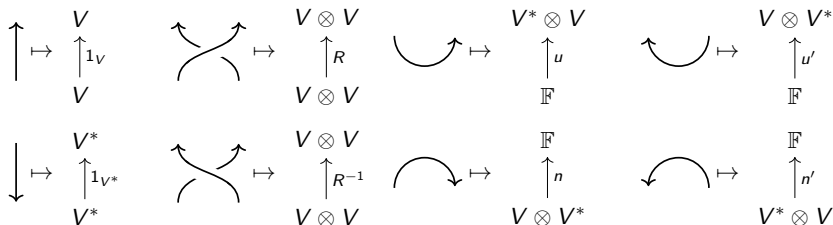
satisfying *some* conditions, e.g. the Yang–Baxter equation:

$$(I \otimes R) \circ (R \otimes I) \circ (I \otimes R) = (R \otimes I) \circ (I \otimes R) \circ (R \otimes I).$$

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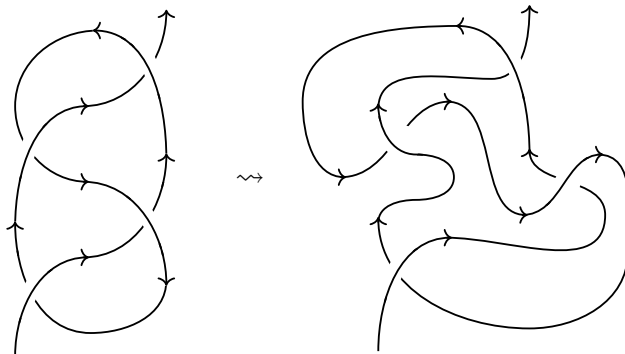


$$u(1) = \sum_i e_i^* \otimes h(e_i), \quad u'(1) = \sum_i e_i \otimes e_i^*,$$

$$n(x \otimes f) = f(h^{-1}(x)), \quad n'(f \otimes x) = f(x)$$

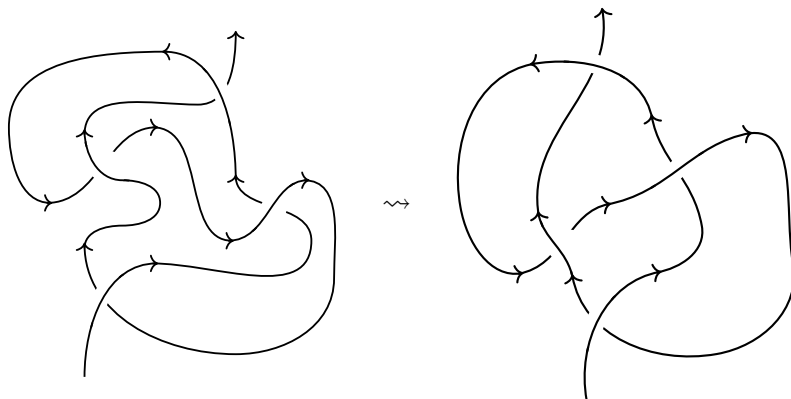
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Example: the 4_1 knot



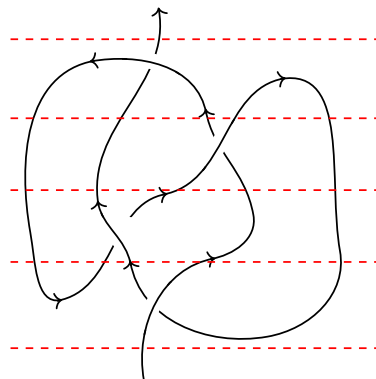
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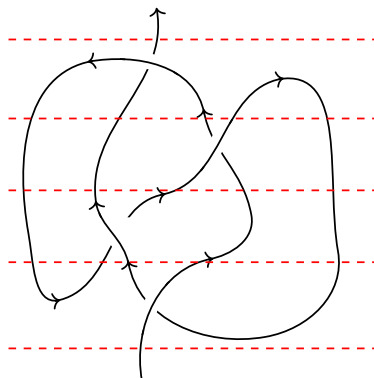
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$$\begin{array}{c}
 V \\
 \uparrow \\
 V^{\otimes 5} \\
 \uparrow \\
 V^{\otimes 5} \\
 \uparrow \\
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 \uparrow \\
 V
 \end{array}
 \quad \mapsto \quad
 \begin{array}{c}
 \left((n' \otimes 1_V) \circ (1_V \otimes R^{-1}) \right) \otimes n \\
 1_V \otimes 1_V \otimes R \otimes 1_V \\
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 u \otimes \left((R \otimes 1_V) \circ (1_V \otimes u') \right)
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Fact

The endomorphism on V we obtain is a scalar multiple of 1_V .
Plugging indeterminants in R and h gives polynomial invariants.

Fix a basis $\mathcal{B} := \{e_i\}$ of V , $R^{\pm 1} \in \text{Aut}(V \otimes V)$ (and h) become matrices whose entries can be denoted by $(R^{\pm 1})_{e_i \otimes e_j}^{e_k \otimes e_l}$ (and $h_{e_i}^{e_j}$).

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$$\sum_{\substack{a_1, \dots, a_{2n-1} \in \mathcal{B} \\ a_0 = a_{2n} = e_1}} \underbrace{(R^{\pm 1})_{a_0 \otimes a_1}^{a_2 \otimes a_3} \cdots (R^{\pm 1})_{a_{2n-3} \otimes a_{2n-2}}^{a_{2n-1} \otimes a_{2n}}}_{\text{a product of length } n},$$

where n is the number of crossings of the knot.

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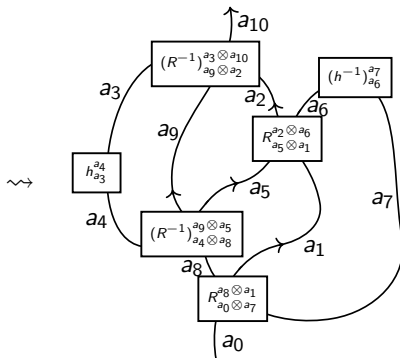
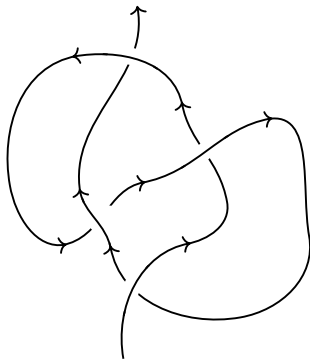
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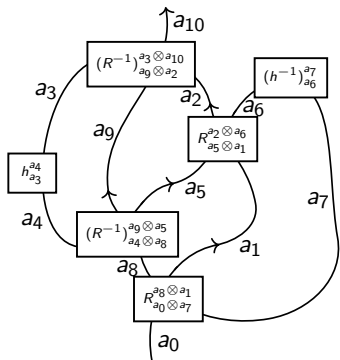
$$n \cdot (\dim V)^{2n-1}$$

multiplications and additions to compute the eigenvalue.

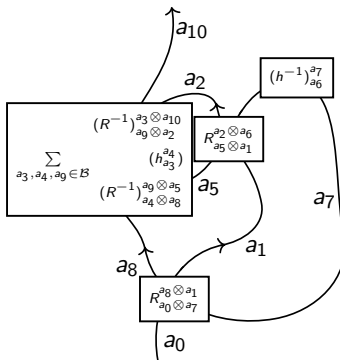
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$\rightsquigarrow$



Definition (Tensor)

An (equi-dimensional) n -tensor T over a ring $\mathcal{R}$ is a tuple $(T_{i_1, \dots, i_n})$ of elements of $\mathcal{R}$ with $i_j = 1, \dots, d$ for each j .

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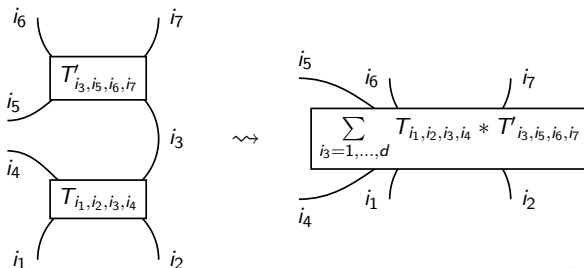
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$$O(c \cdot d^s) \cdot d^{N-2s} = O(c \cdot d^{N-s}),$$

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Definition (Contraction-width)

Let $\mathfrak{S}(\Gamma)$ denote the set of all contraction sequences on Γ :

$$\underbrace{\min_{\mathcal{S} \in \mathfrak{S}(\Gamma)} \left(\overbrace{\max_{i=1, \dots, |\mathcal{S}|} \underbrace{(N_i - s_i)}_{\text{local contraction-width of the } i\text{-th contraction}}}^{\text{contraction-width of } \mathcal{S}} \right)}_{\text{contraction-width of } \Gamma}$$

Theorem (L. 2026)

Computing a k -variable Reshetikhin–Turaev knot polynomial on an n -crossing diagram by contracting with contraction-width w is

$$O((\dim V)^w n^{k+2} (\log n)^3).$$

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Let $w(\Gamma)$ and $tw(\Gamma)$ be the contraction-width and tree-width of Γ ,

$$w(\Gamma) \leq \Delta(\Gamma) \cdot (tw(\Gamma) + 1) = O(\Delta(\Gamma) \cdot \sqrt{n}),$$

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Corollary

Computing Reshetikhin–Turaev knot polynomials are $e^{O(\sqrt{n})}$.

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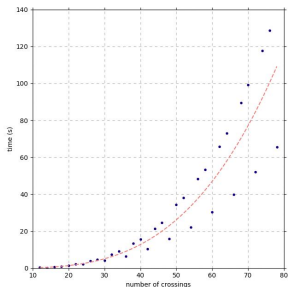
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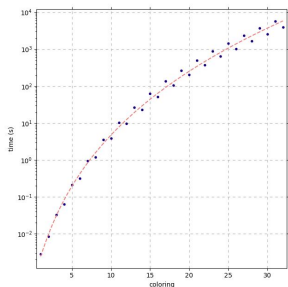
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The 2-colored Links–Gould polynomial for $(3, i)$ -torus links as i ranges from 6 to 39.



The n -colored Jones polynomial for $11n34$ as n ranges from 1 to 32. (plotted in log scale)

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Corollary

If the above conjecture is true, the computation of the 2-colored Links–Gould polynomial gives algorithms for the top two items.

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Definition (Ribbon concordance)

Let J and K be two knots in S^3 , a *concordance* C from J to K is a smoothly embedded annulus $C \subset I \times S^3$ cobounding J and K . C is a *ribbon concordance* if the projection $C \hookrightarrow I \times S^3 \rightarrow I$ is a Morse function with no index-2 critical points.

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Definition (Slice & Ribbon)

A knot is *slice* (resp. *ribbon* or *handle-ribbon*) if there is a concordance (resp. ribbon or handle-ribbon concordance) between it and the unknot.

Restricting to concordances in the standard $I \times S^3$:

$\{\text{ribbon concordances}\} \subset \{\text{handle-ribbon concordances}\} \subset \{\text{concordances}\}$

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Proposition

If any of the following exists, either slice-ribbon or SPC4 is false:

- A handle-ribbon concordant pair (J, K) that is not ribbon;
- A slice knot that is not handle-ribbon;
- A non-slice knot that is handle-ribbon (in a homotopy $I \times S^3$).

Definition (Fibered knots)

A knot K in S^3 is fibered if there exists a one-punctured surface Σ with $\varphi \in \text{Aut}(\Sigma)$ such that the mapping torus of φ is homeomorphic to $S^3 \setminus K$. φ is called the *monodromy* of K .

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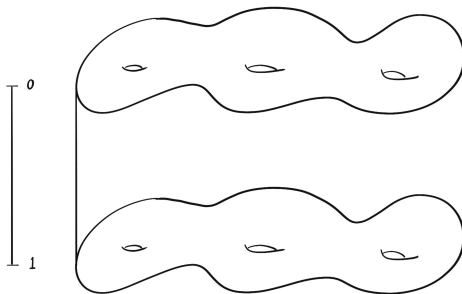
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Definition (Compression body)

A *compression body* M of a surface Σ is a 3-manifold formed from $\Sigma \times I$ by attaching 2- and 3-handles to $\Sigma \times \{1\}$. An automorphism $\varphi \in \text{Aut}(\Sigma)$ compresses in M if φ extends from $\Sigma \cong \Sigma \times \{0\}$ to M .

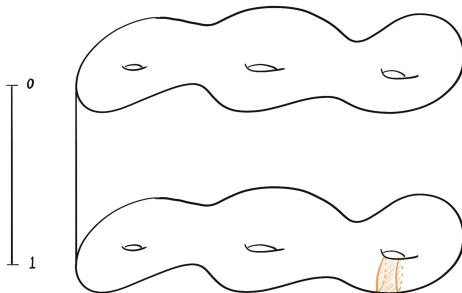
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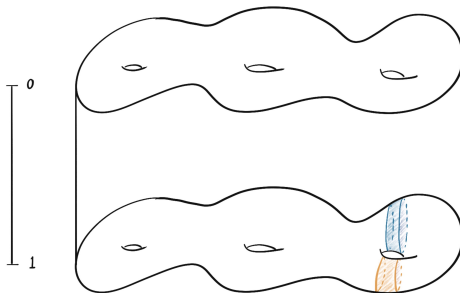
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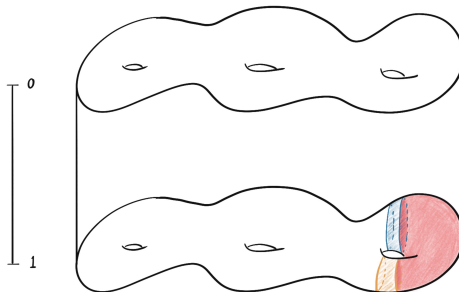
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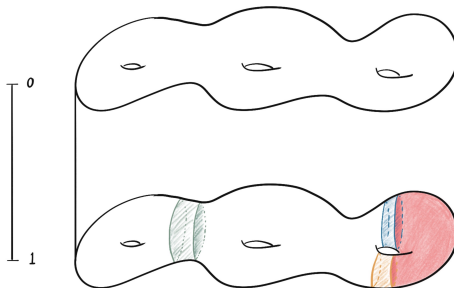
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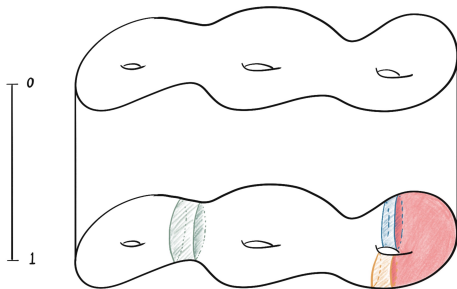
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Observation: compression bodies are determined by curves describing the 2-handle attachments.

Theorem (Casson & Gordon, 1983)

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Theorem (Casson & Long, 1985)

For a pseudo-Anosov surface homeomorphism, there is an algorithm that finds all minimal compression bodies where it compresses.

Let $\mathcal{C}$ be finitely many nonintersecting non-parallel essential curves on Σ and $M_{\mathcal{C}}$ be the minimal compression body where $\mathcal{C}$ bounds.

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Lemma

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Lemma

There is an algorithm that, given two set of curves $\mathcal{C}$ and $\mathcal{C}'$, outputs a (finite) list $\mathcal{L}_{\mathcal{C}, \mathcal{C}'}$ of all minimal compression bodies where every curve in $\mathcal{C} \cup \mathcal{C}'$ bounds a disk.

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In other words, if M is a compression body where $\mathcal{C} \cup \mathcal{C}'$ bounds, then there exists $M' \in \mathcal{L}_{\mathcal{C}, \mathcal{C}'}$ such that $M' \subset M$.

An Iterative Procedure

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- Apply the algorithm mentioned to obtain the list $\mathcal{L}_{\mathcal{C}, \varphi(\mathcal{C})}$.

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Procedure:

- Apply the algorithm mentioned to obtain the list $\mathcal{L}_{\mathcal{C}, \varphi(\mathcal{C})}$.
- Check if any compression body in $\mathcal{L}_{\mathcal{C}, \varphi(\mathcal{C})}$ coincides with $M_{\mathcal{C}}$.
If so, add $M_{\mathcal{C}}$ to the list of answers.

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Otherwise, the defining curves of M' all bound in M , so are their images under φ , hence iterating the above argument gives a strictly increasing sequence deemed to reach M .

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Theorem. (Casson & Long)

If φ is pseudo-Anosov, there exists $\ell > 0$ such that for any compression body M where φ compresses, there exists an essential curve with length less than ℓ that bounds.

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A Slice of Progress

24 knots whose ribbon-ness is unknown:

- The following are handle-ribbon:

$18nh00000601$	$18nh01872227$	$20fibc000040$	$20fibc000336$
$20fibc008969$	$20fibc017417$	$20fibc063973$	

- The following are not handle-ribbon (subject to numerical errors):

$18nh04892103$	$19nh000941086$	$19nh010449461$
$20fibc015896$	$20fibc084041$	$20fibc113742$

- The following compress to disconnected surfaces, warranting further investigation:

$18nh00068958$	$18nh00251375$	$18nh00584034$	$19nh000054314$	$19nh000107587$
$20fibc000246$	$20fibc000392$	$20fibc011793$	$20fibc046185$	$20fibc087197$